ODE Parameter Estimates for Bd in *Litoria aurea* (Green and Golden Bell Frog)

Overview

This document refines the general Bd compartmental model parameters specifically for *Litoria aurea*, an endangered Australian frog and the focal species of the Waddle et al. (2024) "frog sauna" study. *L. aurea* is notably more susceptible to Bd than many other species, with a disease-causing threshold roughly **150× lower** than *Rana muscosa*. This dramatically affects parameter calibration.

β — Disease-induced mortality: *L. aurea* is highly susceptible, especially at cool temperatures

L. aurea has one of the lowest known disease-causing thresholds of any studied amphibian.

Stockwell et al. (2016), FEMS Microbiol. Lett. 363:fnw111 — Determined the species-specific lethal threshold for *L. aurea* via radio-tracking and qPCR: *L. aurea* had a **78% probability of developing chytridiomycosis past a threshold of only 68 zoospore equivalents (ZE) per swab**, with disease occurring across a variable range of 0.5–490 ZE. For comparison, *R. muscosa*/*R. sierrae* populations collapsed at ~10,000 ZE (Vredenburg et al. 2010). This means *L. aurea* is roughly **two orders of magnitude more sensitive** to Bd load than the Sierran species.

Campbell et al. (2019), Sci. Rep. 9:83, DOI: 10.1038/s41598-018-35874-7 — The definitive temperature × mortality experiment on *L. aurea*. Post-metamorphic juveniles (6–10 weeks old) were inoculated with Bd (strain Gibbo River-Llesueuri-00-LB-1, 9.05×10^6 zoospores/mL) and held at either 12°C or 24°C for 48 days:

- **At 12°C: 30% survival** (14/20 died) → $\beta \approx$ **$-\ln(0.30)/48 \approx 0.025$ day^{-1}**
- **At 24°C: 90% survival** (2/20 died) → $\beta \approx$ **$-\ln(0.90)/48 \approx 0.002$ day^{-1}**
- **Uninfected controls: 95% survival** at both temperatures
- Infection loads were significantly higher and increased faster at 12°C than at 24°C despite 24°C being within Bd's optimal growth range in culture

This temperature-dependent mortality is critical: β for *L. aurea* spans roughly an **order of magnitude** between winter and summer conditions in the Sydney region.

Abu Bakar et al. (2016), Oecologia, DOI: 10.1007/s00442-016-3607-4 — Age-structured susceptibility experiment on *L. aurea*, comparing recently metamorphosed juveniles, subadults, and adults over 95 days after Bd exposure. Key findings: susceptibility and infection intensity varied with ontogeny. Juveniles had the fastest infection buildup and lowest immune response

(measured via phytohaemagglutinin skin swelling assay). Adults showed greater immunocompetence and better survival outcomes. This validates incorporating age-structured β if extending the model.

Recommended β for *L. aurea*:

Condition	β (day^{-1})	Source
Winter (~12°C), juveniles	**0.025**	Campbell et al. 2019
Summer (~24°C), juveniles	**0.002**	Campbell et al. 2019
Central estimate (annual average, ~18°C)	**~0.010–0.015**	Interpolated
Adults, enzootic conditions	**~0.002–0.005**	Abu Bakar et al. 2016; Waddle et al. 2024

α — Transmission rate: validated as density-dependent for L. aurea via salinity experiments

No direct estimation of α exists for *L. aurea*, but several experiments provide constraints.

Stockwell et al. (2012), PLoS ONE 7:e36942 — Used *Litoria peronii* (a closely related Litoria species) to establish that Bd transmission and infection load are significantly affected by water salinity. Frogs in 0 ppt had the highest infection loads and lowest survival; 3 ppt NaCl significantly improved survival. This demonstrates that α is modulated by the aquatic environment. The experimental setup (individual frogs in 15L aquaria with 10^7 zoospores) shows transmission occurs rapidly at standard zoospore densities.

Clulow et al. (2018) — Confirmed in outdoor mesocosm experiments with *L. aurea* that Bd transmission between infected and uninfected individuals was significantly reduced at 3.5 ppt salinity compared to freshwater, validating density-dependent transmission in this species.

Waddle et al. (2024), Nature 631:344–349 — Provided indirect transmission data from the mesocosm experiment: frogs in unshaded (hot) greenhouses had roughly **100-fold lower infection intensity** than frogs in shaded (cool) greenhouses. In the reinfection experiment (n=55), naive frogs exposed to Bd in mesocosms had **22% survival** — confirming rapid and efficient transmission in shared aquatic habitat at ecologically relevant densities.

Recommended α for *L. aurea*: Given the very low disease threshold (68 ZE), *L. aurea* likely has an effective α at the higher end of the general range. Use **$\alpha \approx 5\times10^{-4}$ to 5×10^{-3} $frog^{-1}$ day^{-1}** (i.e., a factor of ~5–10× higher effective transmission than values calibrated for *R. muscosa*, reflecting the lower load needed to cause disease). Sensitivity analysis across 10^{-4} to 10^{-2} is appropriate.

γ — Cadaver transmission: no *L. aurea*-specific data; use general estimates

No *L. aurea*-specific cadaver transmission studies exist. The general estimates from the initial report remain applicable: **$\gamma \approx 0.001\alpha$ to 0.01α**. The key Australian-relevant consideration is that *L. aurea* inhabits ephemeral and semi-permanent wetlands where cadavers may desiccate rapidly in summer, effectively increasing δ and reducing the R1 → S transmission window. In permanent ponds, the general estimates apply.

Recommended γ for *L. aurea*: 10^{-6} to 10^{-4} frog^{-1} day^{-1} (scaled to α).

δ — Cadaver Bd decay: general estimates apply, modulated by Australian climate

The Johnson & Speare (2003, 2005) data on Bd persistence in water (3–7 weeks) and moist soil (up to 12 weeks) were conducted in Australia and are directly applicable to *L. aurea* habitat. Sydney's climate (warm summers, mild winters) means:

- **Summer:** Higher temperatures and frequent drying of ephemeral wetland margins accelerate Bd die-off on cadavers. $\delta \approx$ **0.05–0.10 day^{-1}** (10–20 day persistence).
- **Winter:** Cooler, moister conditions extend Bd viability. $\delta \approx$ **0.015–0.030 day^{-1}** (30–70 day persistence).
- **Annual average:** $\delta \approx$ **0.033 day^{-1}** (~30 days).

Recommended δ for *L. aurea*: 0.020–0.050 day^{-1} (central: **0.033 day^{-1}**).

η — Antifungal treatment recovery: general protocol, but note *L. aurea* growth sensitivity

The standard itraconazole protocol (0.0025%, 5 min/day, 6 days) is species-independent and gives **$\eta = 1/6 \approx 0.167$ day^{-1}**. No *L. aurea*-specific itraconazole study has been published, but the closely related species used in the Australian-developed protocols (*L. caerulea*, *L. peronii*) respond well.

The Hardy et al. (2015) finding that itraconazole-treated *R. cascadae* had **22% lower body mass** at 3 weeks is relevant because Campbell et al. (2019) showed *L. aurea* infected with Bd already has reduced fat stores and elevated metabolic rate. Itraconazole treatment might compound this energetic stress in *L. aurea*.

Recommended η for *L. aurea*: 0.167 day^{-1} (6-day protocol) as central estimate. Range: **0.091–0.167 day^{-1}** (6–11 day protocols).

ζ — Frog sauna entry/clearance: *L. aurea*-specific data available (this is the focal species!)

The Waddle et al. (2024) Nature study was conducted entirely on *L. aurea*, making this the best-constrained parameter.

Waddle et al. (2024), Nature 631:344–349 — Key *L. aurea*-specific findings:

1. **Thermal preference:** Infected *L. aurea* selected body temperatures of approximately **30°C** in a thermal gradient; their preferred body temperature was ~29°C.
2. **Clearance time in lab:** "Six days after being moved into the thermal gradient habitat, the 20 frogs tested had an average infection intensity of **zero**." → Clearance rate once inside = **1/6 ≈ 0.167 day^{-1}**.
3. **Shelter use in field:** Frogs were observed in the greenhouse (hotspot) shelters approximately **four times more often than would be expected by chance** based on available habitat area. This gives a behavioral preference multiplier of ~4×.
4. **Mesocosm infection intensity:** Frogs with access to unshaded (hot) shelters had roughly **100-fold lower Bd infection intensity** than those in shaded (cool) shelters.
5. **Acquired immunity:** Pathogen-experienced *L. aurea* had **86% survival** upon re-exposure vs. **22% survival** for naive frogs — a **23× odds ratio**. This is an extraordinary result and strongly argues for a "recovered with immunity" compartment rather than returning treated frogs to S.

Interpreting ζ: In your model, ζ governs the rate at which frogs (both S and I) enter T1 (frog sauna). This combines (a) the rate of encountering a shelter and (b) the probability of using it. From the Waddle data:

- If shelters cover ~5% of habitat and frogs use them 4× preferentially, the effective daily encounter-and-use probability is roughly 4 × 0.05 = 0.20 per day.
- Once inside, clearance takes ~6 days.
- The combined "entry-to-clearance" rate depends on how ζ is structured in your ODE. If ζ represents only entry rate (and frogs stay in T1 for a duration governed by the thermal clearance dynamics), then **ζ ≈ 0.10–0.20 day^{-1}** for *L. aurea* with adequate shelter density.

Recommended ζ for *L. aurea*: 0.10–0.20 day^{-1} (central: **0.167 day^{-1}**, matching the 6-day clearance). This is a control variable — ζ increases with more shelters deployed.

ξ — Rate of entering antifungal treatment: management-dependent control variable

No *L. aurea*-specific field antifungal treatment studies have been published. However, capture-mark-recapture data from *L. aurea* field studies provide capture probability estimates:

Beranek et al. (2021), J. Environ. Manage. 291:112742 — Weekly capture-mark-recapture at constructed wetlands in the Hunter Valley, NSW. Monitored *L. aurea* populations across three breeding seasons. Reported a **3.3-fold population increase** in constructed wetlands. Weekly recapture rates for *L. aurea* are typically in the range of **0.10–0.30 per survey** during the active season (spring/summer), but much lower in winter when frogs are less detectable.

Recommended ξ for *L. aurea*:

Scenario	ξ (day^{-1})	Interpretation
Low intensity (monthly surveys)	**0.005**	~3.5% of infected frogs captured/treated per week
Medium intensity (weekly surveys)	**0.020**	~14% per week
High intensity (active trapping)	**0.040**	~28% per week

Central estimate for a feasible field program: **ξ ≈ 0.02 day^{-1}**.

Consolidated Parameter Table — *Litoria aurea*-specific

Parameter	Symbol	L. aurea range	Central estimate	Units	Key L. aurea source
Transmission rate (live → susceptible)	α	5×10^{-4} to 5×10^{-3}	~1×10^{-3}	$frog^{-1}$ day^{-1}	Stockwell 2016 (low threshold); Clulow 2018
Cadaver transmission rate	γ	10^{-6} to 10^{-4}	~10^{-5}	$frog^{-1}$ day^{-1}	No L. aurea data; scaled from α
Antifungal recovery rate	η	0.091–0.167	0.167	day^{-1}	Standard itraconazole protocol (Brannelly 2012)
Frog sauna entry/clearance rate	ζ	0.10–0.20	0.167	day^{-1}	**Waddle et al. 2024 (direct L. aurea data)**

Parameter	Symbol	L. aurea range	Central estimate	Units	Key L. aurea source
Disease-induced mortality	β	0.002–0.025	0.015	day^{-1}	**Campbell et al. 2019 (direct L. aurea data)**
Cadaver Bd decay rate	δ	0.020–0.050	0.033	day^{-1}	Johnson & Speare 2003, 2005 (Australian data)
Treatment capture rate	ξ	0.005–0.040	0.020	day^{-1}	L. aurea CMR estimates (Beranek 2021)

Key *L. aurea*-Specific Considerations for Model Design

1. The 68 ZE threshold changes everything

The Stockwell et al. (2016) finding that *L. aurea* develops lethal chytridiomycosis at only **68 ZE** (vs. ~10,000 for *R. muscosa*) means this species is operating in a fundamentally different regime. Small increases in zoospore exposure translate rapidly to lethal outcomes. This implies that α should be calibrated to produce population crashes matching observed timelines (typically 1–3 years post-Bd arrival in new *L. aurea* sites; Stockwell et al. 2008).

2. Temperature is the dominant environmental driver

Campbell et al. (2019) showed a **12.5× difference in mortality** between 12°C and 24°C in *L. aurea*. This isn't a minor seasonal effect — it's the difference between population persistence and collapse. The commented-out seasonal forcing term in your ODE ($\sigma \cdot \cos(2\pi t)$) should seriously be considered for *L. aurea*, with σ calibrated to the Sydney temperature regime (winter lows ~8–12°C, summer highs ~24–30°C).

3. Acquired immunity from frog saunas is dramatic

Waddle et al. (2024) showed that *L. aurea* that cleared Bd via thermal treatment had **23× higher odds of surviving re-exposure** (86% vs. 22%). This argues strongly for adding a "recovered-immune" compartment (R_immune) rather than returning frogs from T1 directly to S. Without this, the model will underestimate the long-term benefit of the frog sauna intervention.

4. Age structure matters for *L. aurea*

Abu Bakar et al. (2016) demonstrated clear ontogenetic variation in Bd susceptibility in *L. aurea*: juveniles are most vulnerable, with the lowest immunocompetence (measured by PHA response). If incorporating age structure, consider separate β values for juveniles (β ≈ 0.02–0.025) and adults (β ≈ 0.002–0.005). The Cook et al. (2022) finding from *R. cascadae* — that Bd

prevalence jumps from ~10% to ~70% in untreated metamorphs within 5 weeks — appears to parallel the *L. aurea* juvenile vulnerability pattern.

5. Salinity as an additional environmental modulator

Stockwell et al. (2012, 2015) and Clulow et al. (2018) showed that modest salinity increases (2–4 ppt NaCl) significantly reduce Bd transmission and infection intensity in *L. aurea* and related Litoria species. Since *L. aurea* naturally occupies coastal ecotonal habitats, this could be an additional environmental parameter to incorporate — effectively reducing α in saline conditions.

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